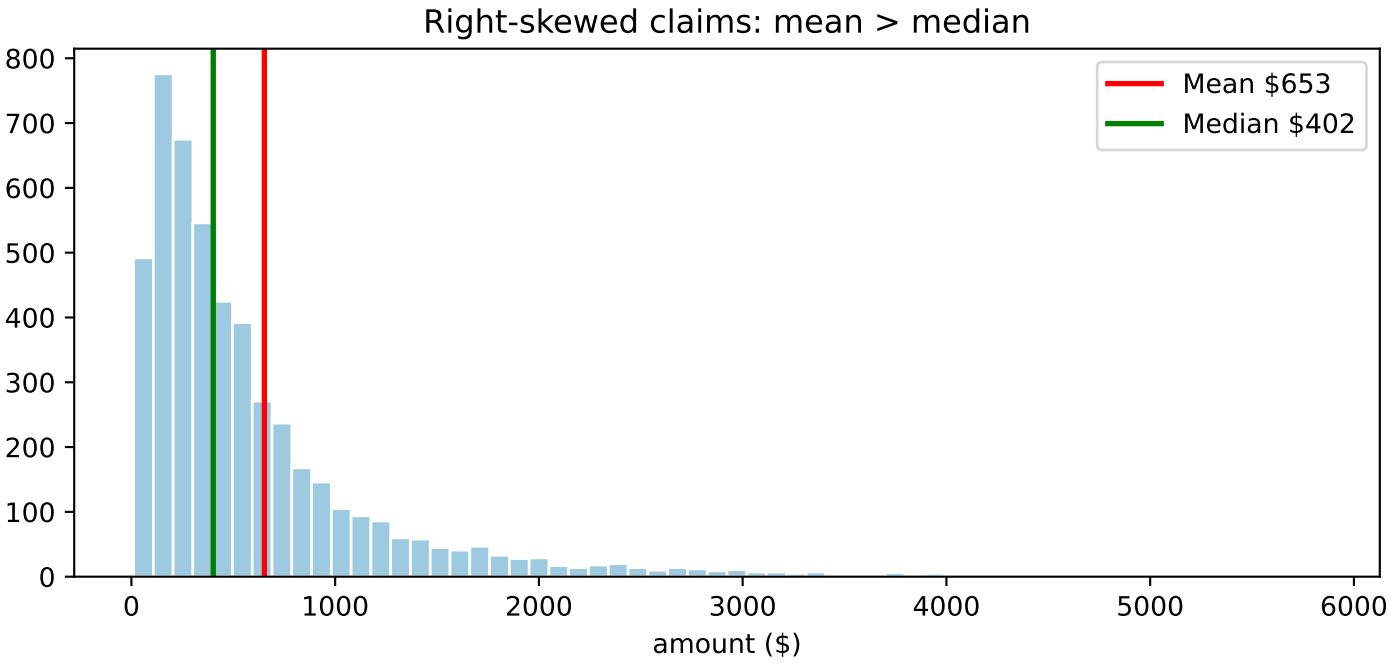
# Day 1 Cheat Sheet: Descriptive Statistics

KEY IDEAS
- Mean: average; pulled toward the long tail by skew
- Median: middle value; robust to outliers
- Std dev: typical spread around the mean
- Percentiles: Q1=25th, Q3=75th; IQR = Q3 - Q1
- Outlier rule: < Q1-1.5xIQR or > Q3+1.5xIQR
- Skewed $? Report median; model log1p(amount)

THIS DATA (n=5000)
mean $653 | median $402 | std $845
Q1 $202 | Q3 $758 | IQR $556 | outliers 408



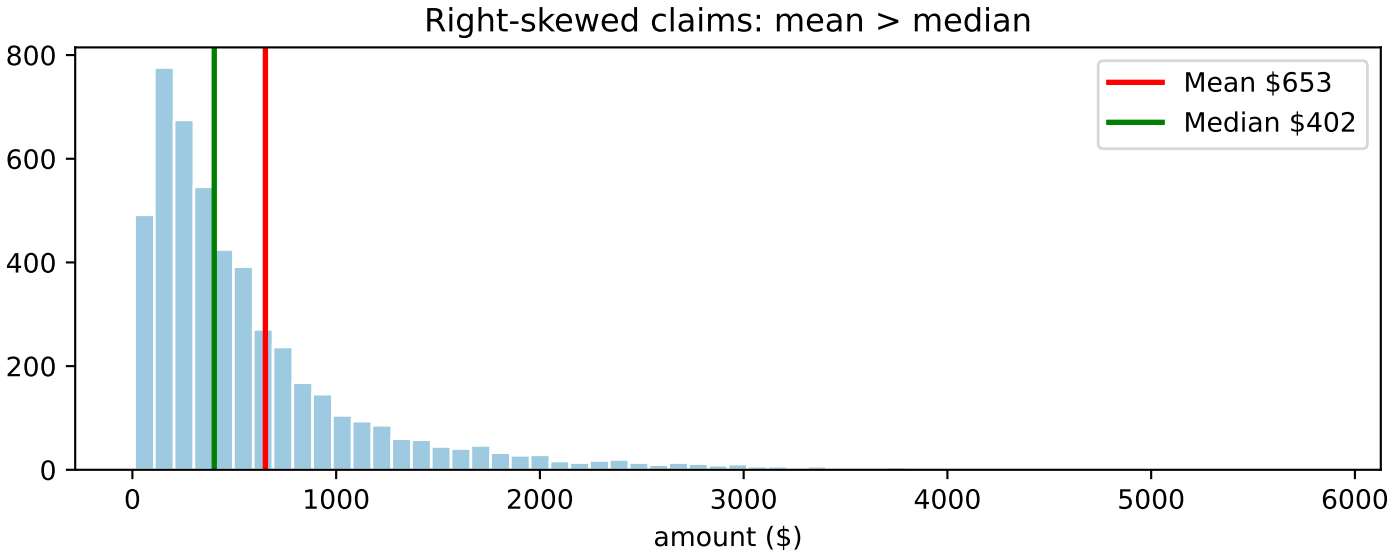

```
df['amount']=np.random.lognormal(mean=6, sigma=1, size=5000)
df.groupby('plan_type')['amount'].describe()
q1,q3=df.amount.quantile([.25,.75]); iqr=q3-q1
out=df[(df.amount<q1-1.5*iqr)|(df.amount>q3+1.5*iqr)]
df['log_amt']=np.log1p(df.amount)
```

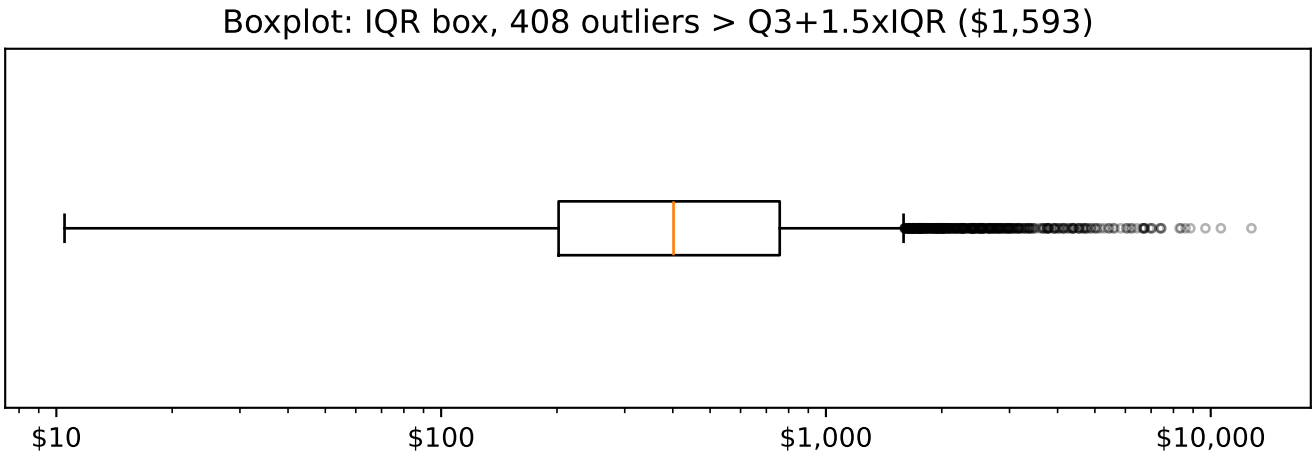
# Infographic: Why the Median Wins on Claims

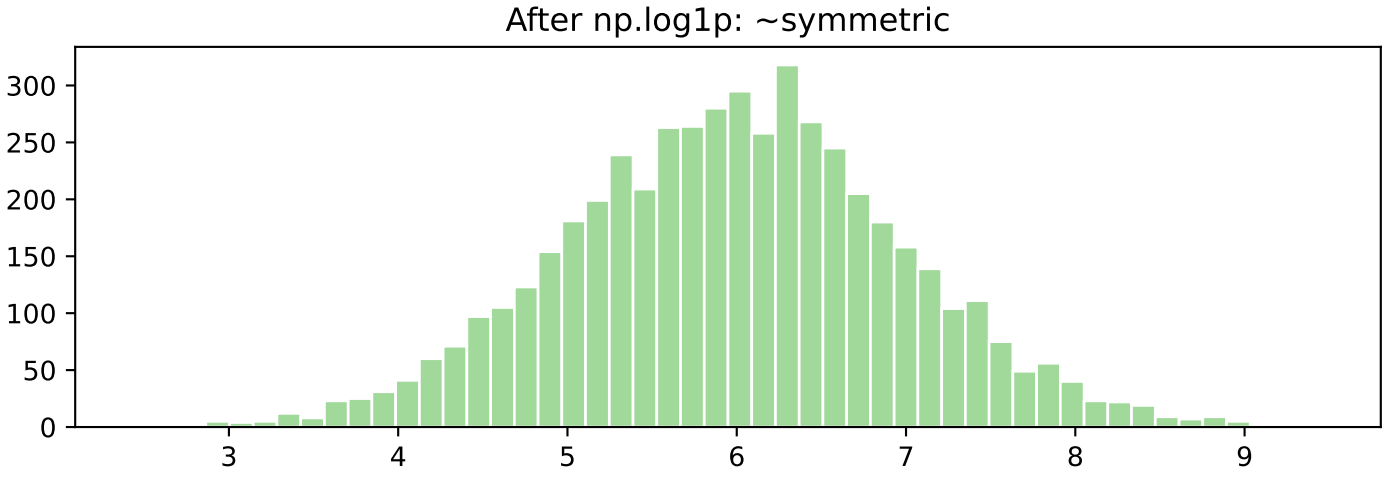
**MEAN**
**$653**

**MEDIAN**
**$402**

**STD**
**$845**



# Hands-On Worksheet: ds-genai-90-days / day01

[ ] Create repo ds-genai-90-days (day01/, data/, README.md)

[ ] conda create -n ds90 python=3.11 pandas numpy matplotlib seaborn scipy jupyter -y

[ ] Simulate amounts: np.random.lognormal(mean=6, sigma=1, size=5000)

[ ] Add member_id and plan_type (HMO/PPO)

[ ] Stats: df.groupby('plan_type')['amount'].describe(), .skew()

[ ] IQR outliers: q1,q3=df.amount.quantile([.25,.75])

[ ] Plot raw vs np.log1p(amount) histograms

[ ] Write 3 stakeholder insights in markdown

[ ] git add . && git commit -m 'Day 1: descriptive stats' && git push

[ ] Stretch: Snowflake MEDIAN, PERCENTILE_CONT(0.75)

```
df['amount']=np.random.lognormal(mean=6, sigma=1, size=5000)
df.groupby('plan_type')['amount'].describe()
q1,q3=df.amount.quantile([.25,.75]); iqr=q3-q1
out=df[(df.amount<q1-1.5*iqr)|(df.amount>q3+1.5*iqr)]
df['log_amt']=np.log1p(df.amount)
```

Notes: _____

_____

# Day 1 Self-Check Quiz

1. In right-skewed data, which is larger: mean or median? Why?

_____

2. Q1=$200, Q3=$900. What is the IQR? Upper outlier fence?

_____

3. Why use np.log1p instead of np.log?

_____

4. Which statistic would you report to a stakeholder for typical claim cost?

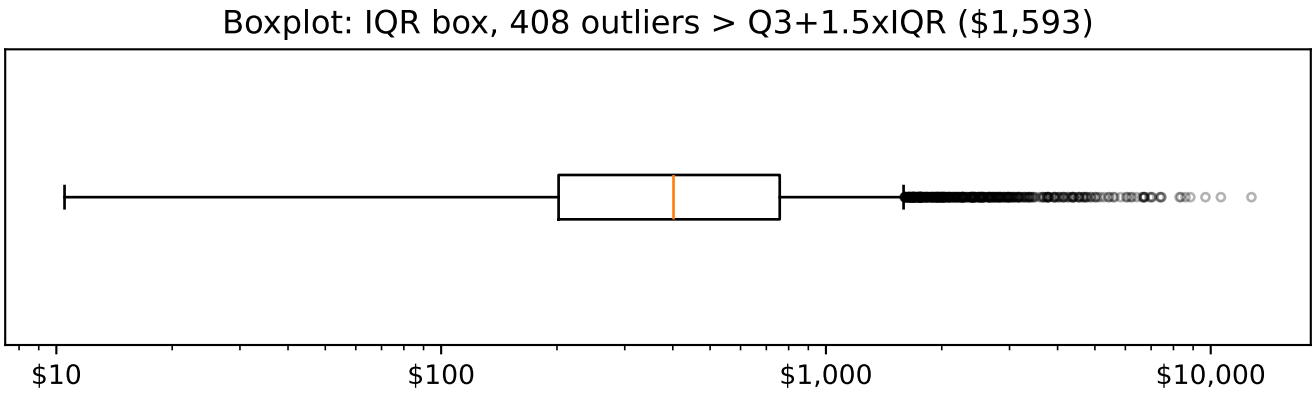
_____

5. What does df.groupby('plan_type')['amount'].describe() return?

_____

6. Does removing outliers change the median much? The mean?

_____




Answers: 1 mean (tail pulls it)  2 IQR $700, fence $1,950  3 handles 0 safely
4 median  5 count/mean/std/min/25/50/75/max per plan  6 median barely; mean a lot